Digital Information: Base Conversions & Two's Complement

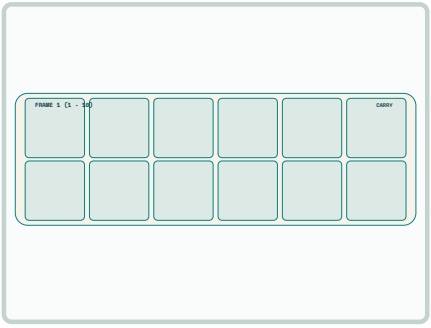
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CSTA.3A-DA-09

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 8 two-tone counters on the Ten-Frame register tray: 1s (white) and 0s (teal) representing byte 0010 1010 (42).

Sketch Model on Paper:
Draw binary place value chart: 128, 64, 32, 16, 8, 4, 2, 1.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: HEX Hexadecimal Logic

1.	HEX	MODE: HEXADECIMAL	Initialize Programme
2.	255	X: 255	Decimal 255 entered
3.	HEX	X: 0xFF	Base convert: 255 DE
4.	BIN	X: 1111 1111	Base convert: FF HEX

EXPECTED LCD X: 0xFF (1111 1111 in BIN)

1 **Convert decimal 42 to hex: [42] [HEX].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Convert hex 1A to decimal: [1][A] [DEC].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Extract low nibble of 0x2A in HEX: 0x2A ÷R 0x10.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Find Two's complement negation of 5 in binary.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: AP CSP Big Idea 2

Digital Information: Base Conversions & Two's Complement • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame 8-Bit Register Studio & Place Value Blocks

CAD Blueprint: ten_frame_tray

Target Device: SC-32 Scientific

OpenSCAD Source File:

Education/scad/ten_frames_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Tray & White/Teal Bit Counters (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Convert decimal 42 to hex: [42] [HEX].** Standard: CSTA.3A-DA-09	0x2A	[4][2] [HEX]
2	**Convert hex 1A to decimal: [1][A] [DEC].** Standard: CSTA.3A-DA-09	26	[HEX] [1A] [DEC]
3	**Extract low nibble of 0x2A in HEX: 0x2A ÷R 0** Standard: CSTA.3A-DA-09	0x0A (10 DEC)	[HEX] [2A] [ENTER] [10] [÷R] [x⇄y]
4	**Find Two's complement negation of 5 in binar** Standard: CSTA.3A-DA-09	-5 (Two's Comp)	[5] [+/-] [BIN]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Remember hex digits A through F represent values 10 through 15 in base 16.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.